

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458257.758 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.126 +/- 0.018
Transit depth (Rp/Rs)^2	1.58 +/- 0.45 [%]
Orbital Inclination (inc)	86.54 +/- 1.23
Airmass coefficient 1 (a1)	1.392 +/- 0.025
Airmass coefficient 2 (a2)	-0.1494 +/- 0.004
Scatter in the residuals of the lightcurve fit is	1.73 %
Best Comparison Star	None
Optimal Aperture	2.82
Optimal Annulus	7.3
Transit Duration (day)	0.09 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.126 ± 0.018 vs 0.1394 (deviation 0.71σ)
Duration fit vs published	0.09 d vs 0.0983 d (deviation 0.8σ)
Mid-time O–C	8.6 min
Depth SNR	6.7
Residual scatter	1.73 %

Light curve

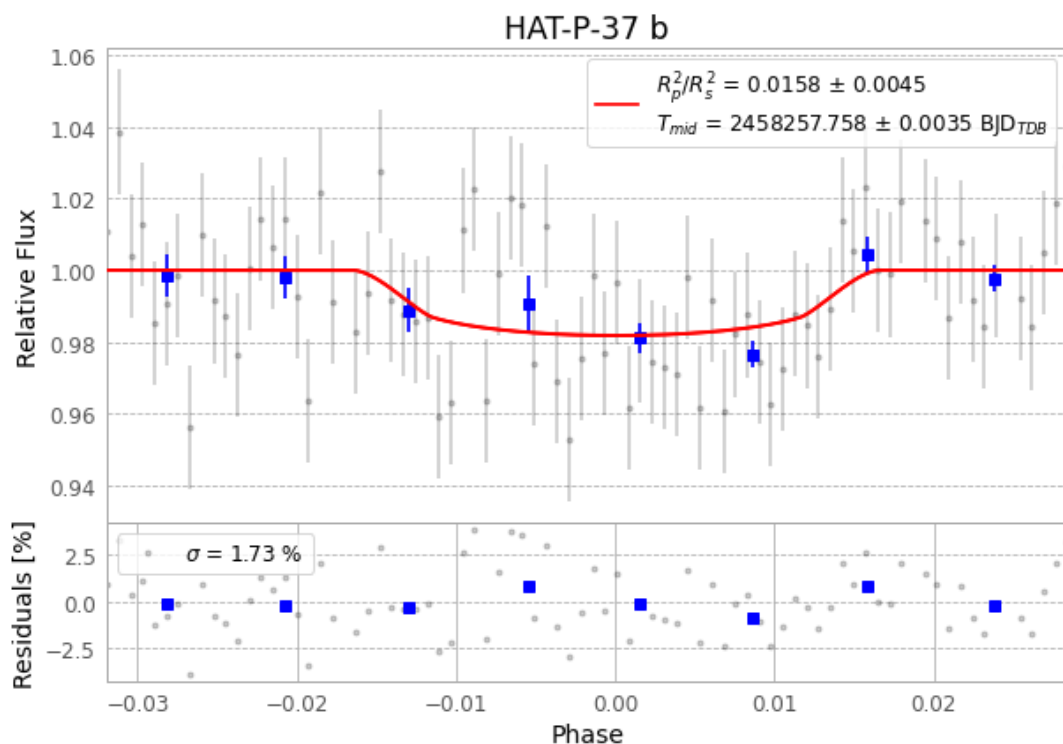


Figure 1 — FinalLightCurve_HAT-P-37 b_2018-05-18.png (SHA-256 a4f1a01f3d905b15...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [375, 183], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e33692d37551a087...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

82 FITS frames (54 MB), HATP-37180519040012.FITS → HATP-37180519080312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-180519.log (cb093fc68be7...), FinalLightCurve_HAT-P-37 b_2018-05-18.png (a4f1a01f3d90...), FinalLightCurve_HAT-P-37 b_2018-05-18.csv (dcc6a2da6a74...)

Compute resources

Wall time 117.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 01:28Z.